



C++ Programming Fundamentals Course Outline

1. Introduction to C++

- History and features of C++
- Structure of a C++ program
- Compilation and execution process
- Basic syntax, keywords, identifiers

2. Input / Output

- cin and cout
- Using manipulators (endl, setw, fixed, setprecision)
- Basic formatting

3. Variables and Data Types

- Built-in data types (int, float, double, char, bool)
- Constants (const, #define)
- Type modifiers (short, long, unsigned, etc.)
- Type casting

4. Operators

- Arithmetic operators
- Relational operators
- Logical operators
- Assignment operators
- Increment / Decrement operators
- Conditional (ternary) operator
- Operator precedence

5. Control Structures

- Conditional statements (if, if-else, switch)
- Looping structures (for, while, do-while)
- Nested loops
- break and continue statements

6. Functions

- Defining and calling functions
- Function parameters (by value, by reference)
- Default arguments
- Inline functions
- Function overloading
- Scope and lifetime of variables

7. Arrays

- 1D arrays (declaration, initialization, traversal)
- 2D arrays (matrices)
- Multi-dimensional arrays (basics)
- Passing arrays to functions

8. Strings

- C-style strings (char arrays)
- String manipulation functions (strlen, strcpy, strcmp, etc.)
- Introduction to string class (basic operations like concatenation, length, etc.)

9. Pointers

- Pointer basics and declaration
- Pointer arithmetic
- Pointers with arrays
- Pointers to pointers
- nullptr
- Pointers with functions

10. Dynamic Memory Management

- new and delete operators
- Dynamic arrays
- Memory leaks and good practices

11. Structures

- Defining structures
- Accessing members
- Array of structures
- Structures with functions
- Nested structures

12. File Handling (Basics)

- Input and output file streams (ifstream, ofstream)
- Reading and writing text files
- File modes (ios::in, ios::out, ios::app)

13. Introduction to Object-Oriented Programming (OOP Basics)

- Concept of classes and objects
- Defining classes and creating objects
- Access specifiers (public, private, protected)
- Constructors and destructors (basic)
- Member functions
- "this" pointer